POSITION PAPER · AI HARDWARE METRICS

# Intelligence Per Milliwatt

*Why Tokens Per Second Is the Wrong Metric and What to Replace It With*

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ABSTRACT

The dominant metric for AI hardware performance is tokens per second — a supply-side measure optimized for cloud serving infrastructure. As inference moves from data centers to edge devices, tokens per second measures the wrong thing. It rewards architectures that excel at batch serving over high-bandwidth memory while ignoring the constraint that actually binds at the edge: power. This paper proposes Intelligence Per Milliwatt (IpMW) as the correct metric for the edge AI era, defines it formally as a composite of reasoning capability, personalization delta, and degradation honesty normalized to average inference power draw, and derives the architectural implications. We argue that IpMW is to edge AI what miles per gallon was to the automobile industry — a demand-side metric that, once standardized, restructures an entire industry's design priorities.

> *This paper is a companion to:* Horvat, J. (2026). *Intelligence at the Edge of the Cloud: A Local Architecture for the AI Era.* Public domain.

SECTION I

## The Wrong Metric

Tokens per second became the dominant AI hardware benchmark because it measures the thing that matters most in cloud serving: throughput. A data center running inference for millions of users needs to move tokens fast. Every millisecond of latency at scale multiplies into real cost. Every token per second of throughput gained is revenue earned or infrastructure cost avoided. The metric is not arbitrary — it was the right measure for the problem that existed.

That problem is not the only problem. As inference moves permanently to edge devices

— smartphones, wearables, embedded systems, physical AI platforms — the constraint that binds is no longer throughput. It is power. A device with 3,000 milliwatts of sustained power budget for AI inference cannot run a model that draws 8,000 milliwatts at peak, regardless of how many tokens per second it produces at that draw. Battery life, thermal envelope, and die area all impose hard limits that tokens per second ignores entirely.

Optimizing for tokens per second at the edge produces the wrong architecture. It rewards large batch processing, high memory bandwidth, and parallel activation — properties that favor server-class hardware. It penalizes the sparse activation patterns, aggressive quantization, and selective expert loading that make edge inference efficient. A chip that delivers 200 tokens per second at 4,000 milliwatts scores lower than one delivering 180 tokens per second at 800 milliwatts, even though the second chip delivers far more value per unit of energy spent.

> *Tokens per second is a supply-side metric. It measures what the hardware produces. The edge AI era needs a demand-side metric — one that measures what the user receives per watt spent.*

→ [1, 2]

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SECTION II

## Supply-Side vs Demand-Side Thinking

The distinction between supply-side and demand-side metrics matters more than it might appear. Supply-side metrics measure what a system can produce under favorable conditions — peak throughput, maximum bandwidth, theoretical TOPS. They are useful for comparing hardware capabilities in isolation. They are poor predictors of user value in deployment.

Demand-side metrics measure what the user actually receives per unit of resource consumed. Miles per gallon is the canonical example. Internal combustion engines were optimized for decades on horsepower — a supply-side metric measuring what the engine could produce. When fuel economy standards forced the industry to optimize on miles per gallon — a demand-side metric measuring value delivered per gallon consumed — the entire architecture of engine design changed. Direct injection, variable valve timing, cylinder deactivation, turbocharging for efficiency rather than power — all of these emerged from optimizing for the right metric.

The AI hardware industry is at an analogous inflection point. The data center era optimized on supply-side metrics because cloud economics rewarded throughput. The edge era changes the economics fundamentally. The user cares about whether the

device is intelligent, whether it knows them specifically, and whether it lasts through a full day of use. None of those outcomes are predicted by tokens per second.

Intelligence Per Milliwatt is the miles per gallon of edge AI. It measures value delivered — intelligent, personalized, honest responses — per unit of energy consumed. Optimizing for it produces different architectural choices than optimizing for throughput, and those choices are the right ones for the edge deployment context.

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SECTION III

## Defining Intelligence Per Milliwatt

A useful metric requires a precise definition. IpMW has two components: a numerator that measures intelligence delivered and a denominator that measures energy consumed. Both require careful specification.

DEFINITION · INTELLIGENCE PER MILLIWATT — FORMAL SPECIFICATION

$$\text{IpMW} = \frac{\alpha \cdot R + \beta \cdot P\Delta + \gamma \cdot DH}{\text{AvgPower}_{mW}}$$

| | |
|---|---|
| R — Reasoning | Normalized score on a reasoning benchmark suite (MMLU, ARC-Challenge, or equivalent), expressed as a percentage of a reference cloud baseline scored at 1.0. Measured on tasks within the device's declared expert coverage domain. |
| PΔ — Personalization Delta | Percentage improvement of the personal adapter model over the base model on a suite of user-specific prompts drawn from actual interaction history. Captures the value added by personal calibration beyond the general capability baseline. |
| DH — Degradation Honesty | Accuracy of the coverage score as a predictor of actual task performance in low-coverage domains. Measured as 1 − (mean absolute error between stated coverage and observed accuracy). A system that hallucinates confidently in low-coverage domains scores near zero regardless of raw capability. |
| α, β, γ — Weights | Application-specific weighting coefficients summing to 1.0. Physical AI deployments weight DH highest. Personal assistant deployments weight PΔ highest. General-purpose deployments may weight equally. Published benchmarks should specify weights used. |
| $\text{AvgPower}_{mW}$ | Average system power draw in milliwatts during a complete inference cycle — from query receipt through response generation — measured at the device level. Includes NPU, memory access, enclave overhead, and CPU coordination. Not peak draw. Not NPU draw alone. Full system average. |

**Measurement note:** Power must be measured at the full system level, not component level. NPU peak draw systematically understates true inference cost by 40–60% due to memory bandwidth, CPU coordination, and thermal management overhead. IpMW benchmarks that report NPU-only power are not comparable to those reporting full system power and should be clearly

distinguished.

The three-component numerator is deliberate. A metric that measured only reasoning capability would reward large models with high benchmark scores regardless of whether they are honest about their limits or calibrated to the user. A metric that measured only personalization delta would reward systems that over-fit to user history at the expense of general capability. Degradation honesty is the component most absent from existing edge AI benchmarks and most consequential for real deployment — a system that confidently hallucinates is actively harmful regardless of its average accuracy.

→ [3, 4]

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SECTION IV

What the Numerator Actually Measures

## *Reasoning Capability (R)*

Normalized against a reference cloud baseline, R measures whether the edge device can actually think — not just generate tokens, but produce responses that reflect genuine reasoning about the query. The normalization against a cloud baseline matters: it grounds the score in something meaningful rather than an arbitrary point scale. A device scoring R = 0.75 delivers reasoning quality at 75% of a capable cloud model on covered domains. That is a concrete, interpretable claim.

Importantly, R is measured only within the device's declared expert coverage domain. A device that covers quantum computing but not genomics should not be penalized on genomics tasks — it should score on what it claims to cover. This connects R to the coverage score: a device with an accurate coverage score and high R within that coverage is genuinely more useful than a device with high R but poor coverage prediction.

## *Personalization Delta (PΔ)*

PΔ captures the value added by personal calibration. A device running only the base model has PΔ = 0. As the personal adapter learns the user's patterns, preferred reasoning styles, domain knowledge, and communication preferences, PΔ rises. A device with PΔ = 0.25 delivers responses 25% more useful on user-specific tasks than the same base model without calibration.

This component is what makes IpMW a metric for personal AI rather than just efficient AI. Two devices with identical R scores can have very different real-world value if one knows the user and the other does not. PΔ makes that difference visible in the

benchmark.

### *Degradation Honesty (DH)*

DH is the most novel component and deserves the most careful definition. It measures the calibration of the system's uncertainty — specifically, how accurately the stated coverage score predicts actual performance when the system operates outside its well-covered domains.

A system that scores DH = 1.0 is perfectly calibrated: when it says 80% coverage, it achieves 80% accuracy on those tasks. A system that says 95% coverage but achieves 60% accuracy, or says 40% coverage but achieves 85% accuracy, scores poorly on DH. The goal is not maximum confidence — it is accurate confidence. A system that honestly says "I don't know" is more useful than one that confidently gives a wrong answer, and IpMW rewards that honesty explicitly.

→ [3, 5]

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SECTION V

## Architectural Implications

IpMW is not just a measurement tool. It is a design objective. The architectural choices that maximize IpMW are different from those that maximize tokens per second, and those differences tell us what edge AI hardware should look like.

DIAGRAM 01 · ARCHITECTURAL OPTIMIZATION MAP — DESIGN CHOICES AND THEIR EFFECT ON IPMW

↑↑ IpMW **Sparse MoE activation — activate 10–20% of experts per query**

Reduces AvgPower dramatically while preserving R within covered domains. Sparse activation is the single highest-leverage architectural choice for IpMW.

↑↑ IpMW **Usage-profile quantization — high precision for frequent experts, aggressive INT4 for fallbacks**

Reduces memory bandwidth and power for rarely-used experts while preserving quality in the user's actual domain profile. Improves both numerator and denominator.

↑ IpMW **Behavioral adapter training — no supervised loss, no external gradient signal**

Incremental on-device adapter updates have near-zero inference-time cost once trained. PΔ rises continuously without increasing AvgPower during inference.

↑ IpMW **Coverage score computation — lightweight router confidence aggregation**

A well-calibrated coverage score improves DH at minimal compute cost. The coverage computation itself adds only a few percent to inference overhead while substantially improving DH.

↑ IpMW **Fixed frozen interpreter head — non-trainable cross-attention, compiled at device build time**

A frozen interpreter head eliminates training-time compute overhead. Its inference cost is bounded and predictable, contributing to AvgPower stability.

↓ IpMW **Full model activation on every query**

Activates all parameters regardless of query domain. Increases AvgPower without proportional increase in R for domain-specific tasks. Tokens-per-second optimization leads here.

↓ IpMW **Uniform precision across all weights**

Wastes memory bandwidth and power maintaining high precision for rarely-accessed parameters. Increases AvgPower without benefit to the user's actual query profile.

↓↓ IpMW **Overconfident responses in low-coverage domains**

Destroys DH score regardless of R quality. A system that hallucinates confidently has IpMW near zero on affected queries. The worst possible outcome for the metric and for the user.

~ IpMW **Hardware secure enclave overhead**

Adds 5–15% to AvgPower depending on implementation. Reduces IpMW slightly. Accepted as a necessary cost for the privacy guarantee — not optimized away.

The optimization map reveals something important: the architectural choices that maximize IpMW are precisely the choices that define the broadcast sync architecture described in the companion paper. Sparse MoE world models, usage-profile quantization, behavioral adapter training, coverage score computation, and a frozen interpreter head are not incidental features — they are the design choices that emerge from optimizing for the right metric.

This is not a coincidence. The architecture was designed for the edge deployment context, and IpMW is the metric that captures that context. Tokens per second would have produced a different architecture. The metric defines the design space.

→ [1, 2, 3, 6]

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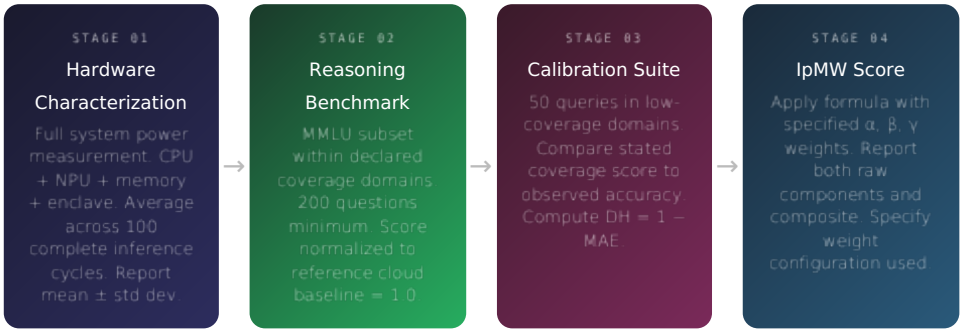
SECTION VI

## Benchmark Methodology

A metric is only as useful as its measurement methodology. IpMW requires a standardized benchmark suite to be comparable across devices and architectures. The following outlines what that suite should contain.

DIAGRAM 02 · IPMW BENCHMARK PIPELINE FOUR-STAGE MEASUREMENT METHODOLOGY




**Reference hardware:** Benchmark results must specify the reference platform (device model, chip generation, OS version, thermal state) and the weight configuration (α, β, γ) used. Results from different weight configurations are not directly comparable. The personalization delta (PΔ) requires a calibrated device — specify number of interaction hours used for adapter training before measurement.

The benchmark has one deliberate complexity: PΔ requires a calibrated device. A freshly deployed device with empty adapters has PΔ = 0 by definition. Published benchmarks should specify the calibration state — typically 10, 50, or 100 hours of interaction — and report PΔ at each milestone. This makes the benchmark more complex but more honest: it reveals how quickly the device becomes useful for a specific user, not just how capable it is out of the box.

Reference hardware for normalization should be updated annually as the frontier advances. The 2026 reference baseline is a capable cloud model at full precision. As edge devices approach cloud parity in covered domains, R scores will naturally rise. The metric scales with the frontier rather than becoming obsolete as hardware improves.

→ [3, 4, 5, 8]

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SECTION VII

## The Metric Defines the Era

Metrics are not neutral. They are design forces. The metric an industry optimizes for determines what gets built, what gets funded, and what gets rewarded. Tokens per second built data centers. TOPS built NPUs. Neither metric built the device that knows you, runs offline, and lasts all day. Intelligence per milliwatt builds that device.

The historical precedent is instructive. Before fuel economy standards, automobile engines were optimized for horsepower because that was what manufacturers competed on and what buyers nominally wanted. The introduction of miles per gallon as a regulatory and consumer metric forced a rethinking of engine architecture from first principles. Technologies that had existed for decades became economically compelling overnight because the metric now rewarded them. The same dynamic is available to edge AI hardware.

In 2026, edge AI chips are already competing on efficiency — the industry consensus is that efficiency is king, and the winners will deliver the most AI per joule. But without a standardized metric that captures the full picture of what "intelligence" means at the edge — reasoning, personalization, and honest uncertainty — competition optimizes for partial proxies. TOPS is a partial proxy. Tokens per second is a partial proxy. IpMW is not a partial proxy. It measures what the user actually receives.

Whoever defines this metric first, and builds hardware that wins on it, defines the next generation of edge AI silicon. The metric creates the battleground. The battleground creates the winners. The winners shape what AI looks like for the next decade of devices.

> *The question is not whether AI will run at the edge. It will. The question is which metric defines what "better" means when it gets there. That question is still open. This paper proposes an answer.*

→ [1, 2, 6, 7]

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APPENDIX

## Tokens Per Second vs Intelligence Per Milliwatt

FULL METRIC COMPARISON — TOKENS/SEC VS IPMW ACROSS KEY PROPERTIES

| PROPERTY | TOKENS PER SECOND | INTELLIGENCE PER MILLIWATT |
|---|---|---|
| Metric type | Supply-side Measures hardware output | Demand-side Measures user value received |
| Optimized for | Cloud Data center batch serving | Edge On-device personal inference |
| Power sensitivity | None — does not account for power draw | Central — power is the denominator |
| Personalization | Invisible — a generic model and a calibrated one score identically | Explicit — PΔ component captures personal calibration value |
| Uncertainty honesty | Invisible — hallucination and accurate response score the same | Explicit — DH component penalizes overconfident hallucination |
| Architectural reward | Full model activation, high memory bandwidth, large batch size | Sparse MoE, usage-profile quantization, behavioral adaptation |
| Binding constraint | Compute throughput and memory bandwidth | Power envelope and thermal budget |
| Measurement scope | Single inference pass, often NPU | Full system across complete |

| | | |
|---|---|---|
| | or GPU only | inference cycle including enclave |
| Scales with frontier | Requires recalibration as model sizes grow | Normalized to reference baseline — scales naturally |
| Analogy | Horsepower — measures engine output | Miles per gallon — measures value delivered per resource consumed |

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